
MiniDog: Dog-Breed Text-to-Image Flow Matching, from Pretraining to Fine-Tuning, on 4 Consumer GPUs

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Abstract. Training flow-matching (FM) transformers usually requires industrial-scale compute, and thus has very little, if any, empirical exercise for student learning despite its vital significance in the AI industry. To fill this gap, this resource provides a runnable pipeline which pretrains a working text-to-image (T2I) model to an FID of 9.4 in just 75 minutes, and fine-tunes it in 20 more minutes, on hardware any student can rent. The recipe follows from a systematic comparison of image tokenizers, representation-alignment losses, and maximum prompt lengths. No pretrained weights are used. Students can run the recipe to do training, evaluation, and visualization themselves. Code, configs and the tutorial notebook are at [MiniDog](#), and data at [HuggingFace](#).

95 minutes total training on 4×RTX 3090 (24GB) → FID = 9.4
75 minutes pretraining (Task 2.1) + 20min supervised fine-tuning (Task 2.2).

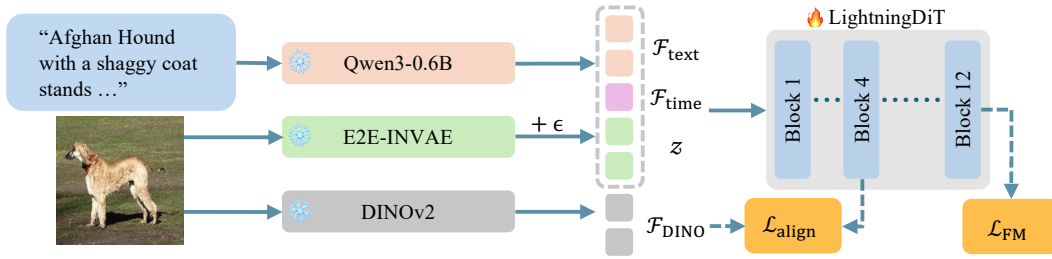


Figure 1: T2I training architecture (❄ = frozen, 🔥 = trained). We use a light Qwen [1] model and DINOv2-B [2] to extract text embedding $\mathcal{F}_{\text{text}}$ and image embedding $\mathcal{F}_{\text{DINO}}$. We use E2E-INVAE [3] as the latent space. Noise ϵ is added to the VAE latent. LightningDiT [4] is adopted as backbone. REPA [5] loss ($\mathcal{L}_{\text{align}}$) and flow-matching loss [6] ($\mathcal{L}_{\text{FM}}$) are used. Timestep t is embedded into $\mathcal{F}_{\text{time}}$.

Concept. The material teaches 1) flow matching basics [6, 7] by training on 2D data using CPUs and 2) pretraining and fine-tuning of text-to-image models using dog images and consumer GPUs.

Leveling and prerequisite knowledge. This material, designed for advanced undergraduate and graduate teaching, requires knowledge of linear algebra, calculus, probability, and deep learning.

Learning objectives and outcomes

- By Task 1, understand FM basics and train FM models on 2D toy data using CPU.
- By Task 2, understand text-to-image flow-matching models and train a MiniDog text-to-image model on consumer GPUs in 95 minutes.
- Understand and compare different choices in architecture and training strategy, including prediction targets, hidden width, representation alignment [5], and tokenizers.
- Understand supervised fine-tuning and score generated images with human-preference models.

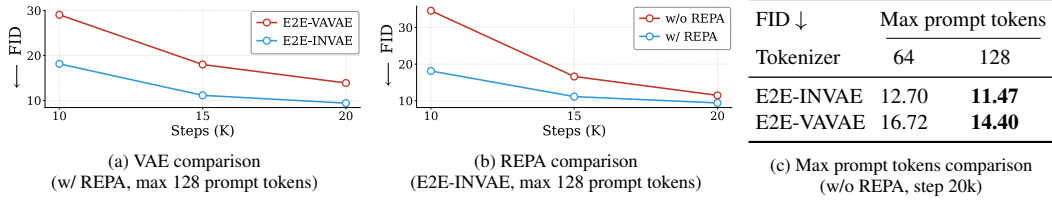


Figure 2: Pretraining FID (lower is better). All runs train for 200 epochs ($\sim 20k$). (a) The E2E-INVAE tokenizer consistently outperforms E2E-VAAE. (b) REPA speeds up convergence and improves the final FID. (c) Longer text prompts (max 128 vs. 64) improve FID for both tokenizers.

Teaching materials summary

Task 1: Flow-matching basics. We provide a self-contained notebook that teaches students how flow-matching models are trained and sampled on CPU. Data is a 2D toy distribution of concentric circles. We embed it in an ambient space of dimension $D = 2$ or $D = 32$. Students train a small MLP on this data with either of two prediction targets, x-prediction and v-prediction [8]. They then inspect the generated samples. Both targets recover the data at $D = 2$. At $D = 32$, v-prediction degrades while x-prediction holds. A second experiment sweeps the hidden width of the v-prediction model at $D = 32$. This tests whether extra capacity rescues it.

Task 2: Pretraining and fine-tuning a text-to-image DiT. The remainder of the resource builds the full T2I pipeline on dog images on consumer GPUs. For pre-training (Task 2.1), we use 26,000 recaptioned dog images from 20 breeds from ImageNet [9]. We make efficient choices of the tokenizer, model architecture, and training recipe so that the T2I model can be pretrained within 75 minutes on 4x3090 GPUs. We show example pretraining FID curves in Fig. 2. Students are expected to reproduce and diagnose the results.

For supervised fine-tuning (SFT, Task 2.2), we generate 2,000 high-quality dog images using FLUX.1-schnell [10] and text prompts. SFT takes 20 minutes on 4x3090s. This resource supports FID [11], human-preference models, and other existing metrics for evaluation. For preference models, we use HPSv2 [12] and PickScore [13], both trained to serve as proxies for human preferences. We report the average score over 500 held-out generated images. From Fig. 3, we clearly observe that the SFT model is superior to the pretrained model. Cherry-picked images and an example prompt are shown.

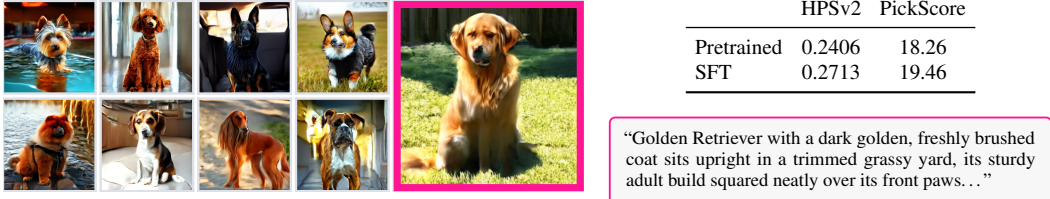


Figure 3: Sample generations from the SFT model (left), with human-preference model scores and the text prompt for the highlighted sample (right). All the other prompts are provided in the linked HuggingFace dataset.

Linked papers

- [1] An Yang et al. Qwen3 technical report. *arXiv:2505.09388*, 2025.
- [2] Maxime Oquab et al. Dinov2: Learning robust visual features without supervision. *arXiv:2304.07193*, 2023.
- [3] Xingjian Leng et al. Repa-e: Unlocking vae for end-to-end tuning with latent diffusion transformers. In *ICCV*, 2025.
- [4] Jingfeng Yao et al. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. In *CVPR*, 2025.
- [5] Sihyun Yu et al. Representation alignment for generation: Training diffusion transformers is easier than you think. In *ICLR*, 2025.
- [6] Yaron Lipman et al. Flow matching for generative modeling. *arXiv:2210.02747*, 2022.
- [7] Xingchao Liu et al. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *ICLR*, 2023.
- [8] Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv:2511.13720*, 2025.
- [9] Olga Russakovsky et al. Imagenet large scale visual recognition challenge. *IJCV*, 2015.
- [10] Black Forest Labs. Flux. <https://github.com/black-forest-labs/flux>, 2024.
- [11] Martin Heusel et al. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *NeurIPS*, 2017.
- [12] Xiaoshi Wu et al. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv:2306.09341*, 2023.
- [13] Yuval Kirstain et al. Pick-a-pic: An open dataset of user preferences for text-to-image generation. *NeurIPS*, 2023.